Project Presentation of Copyright IOHE (22CS422)

On

CaloryInsight

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Table of Content

1. Introduction

2. Problem statement

3. Objective of the Project

4. Existing System

5. Proposed System

7. Data Flow Diagram

6. Technologies Used
    6.1 Mern Architecture

8. Agile Methodology

10. Key Findings Or Outcomes
    10.1 Outcome 1
    10.2 Outcome 2
    10.3 Outcome 3

9. System Architecture

11. Conclusion & Future Enhancements

12. References

Introduction

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•In today's busy lifestyle, many people find it difficult to maintain a healthy routine and keep track of their daily food intake, exercise, and overall fitness.

•CaloryInsight is a smart health and nutrition tracking system developed to help users monitor their calories, check BMI, and make better food and fitness decisions.

•The platform provides useful tools like Calorie Burn Calculator, BMI Calculator, Food Calorie Finder, Goal Tracking System, and Daily Health Logs in one place.

•It allows users to easily understand their body health, set personal fitness goals, and track their daily progress through a simple and user-friendly interface. The project is designed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), which ensures smooth performance, fast processing, and secure data management.

•The main aim of CaloryInsight is to promote health awareness and encourage people to adopt a healthier lifestyle using modern web technology.

Problem statement

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•In this modern era , many people ignore their health due to busy schedules and lack of proper time for fitness monitoring.

•Most users do not have a simple and reliable platform where they can track calories, check BMI, and monitor their daily food intake in one place.

•Existing health applications are often complicated, expensive, or provide only limited features, making them difficult for regular users to use effectively.

•People often make unhealthy food choices because they are unaware of the nutritional value and calorie content of the food they consume.

•There is a need for a user-friendly and complete health management system that helps users maintain fitness goals, track progress, and make better health decisions.

Objective of Project

The main objective of this project is to develop a smart and user-friendly health management system that helps users maintain a healthy lifestyle. The system aims to provide an easy way for users to calculate calories burned during different physical activities and exercises. It helps users check their Body Mass Index (BMI) and understand their health condition based on standard BMI categories. The project also allows users to search for food items and view their nutritional information such as calories, proteins, fats, and carbohydrates. Another objective is to help users set personal fitness goals, maintain daily health records, and track their progress over time. The overall goal of CaloryInsight is to promote health awareness and support better decision-making for fitness and nutrition using modern web technology.

Existing System

In the existing system, most health-related tools such as calorie calculators, BMI checkers, and food nutrition trackers are available on separate platforms, which makes the process time-consuming for users. Many existing applications are complex to use and do not provide a simple and user-friendly interface for regular users. Some platforms only offer limited features and require paid subscriptions to access complete health tracking services.

Drawbacks of Existing Systems:-

- Health tools are available on different platforms separately.
- Many applications are complex and difficult to use.
- Some apps provide limited features or require payment.
- No proper system for daily health record tracking.
- Goal tracking and report generation are often missing.

Proposed System

The proposed system, CaloryInsight, provides all health management tools on a single platform for better convenience and easy access. It allows users to calculate calories burned, check BMI, and search food nutrition details in a simple and user-friendly way. The system helps users maintain daily health records and track their fitness progress regularly. Users can set personal fitness goals and monitor their improvement through a personalized dashboard. It also provides report generation and secure login features, making the system more reliable, organized, and effective for long-term health management.

Proposed Interfaces:-

Calorie Burn Calculator

Calculator Food Calorie and Nutrition Finder

Personal Dashboard for Progress Tracking

Fitness Goal Setting and Monitoring

Daily Health Log Management

BMI (Body Mass Index)

Daily Health Log Management

Report Generation (PDF/Text Export)

Technologies Used..

- **MongoDB**

A NoSQL database that stores data in JSON-like documents, providing flexibility and scalability.

- **Express.js**

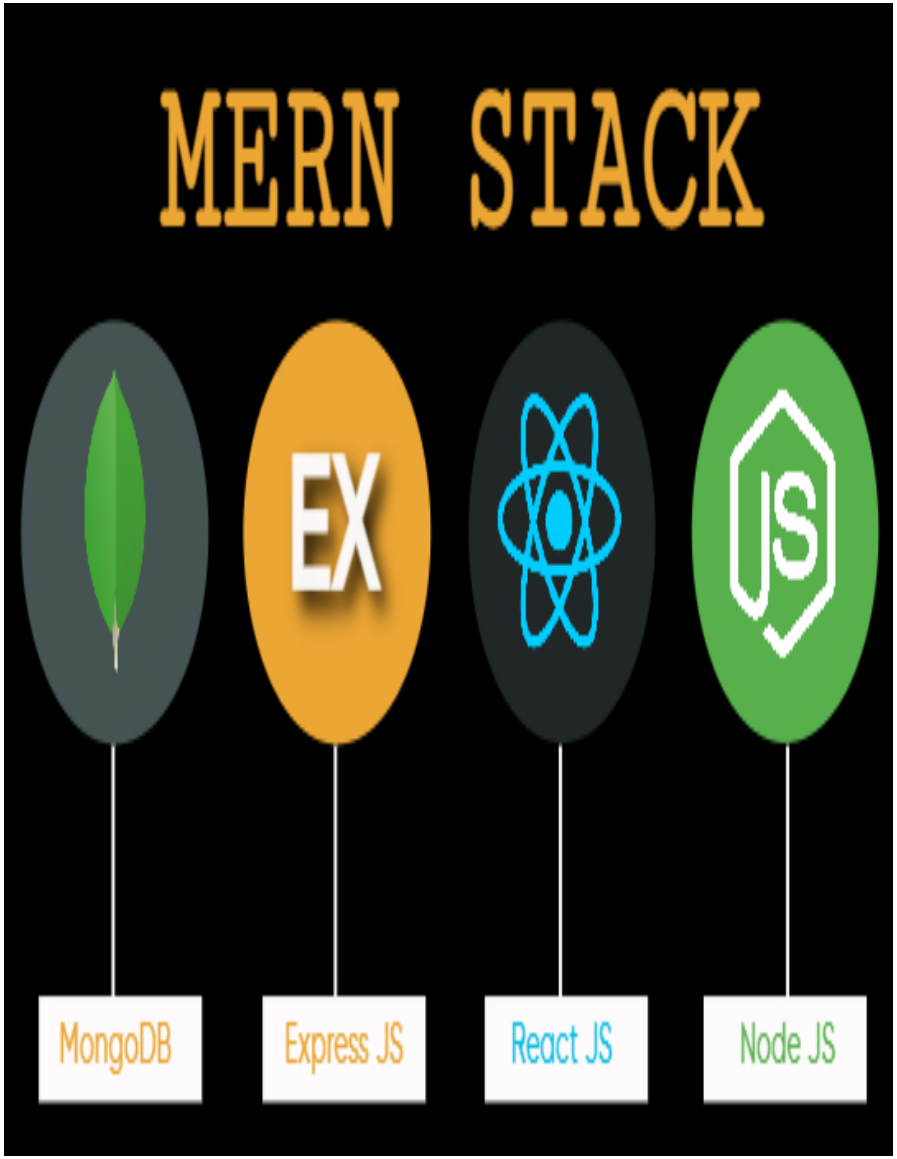
A framework for building web applications using Node.js, handling routing and middleware.

- **React.js**

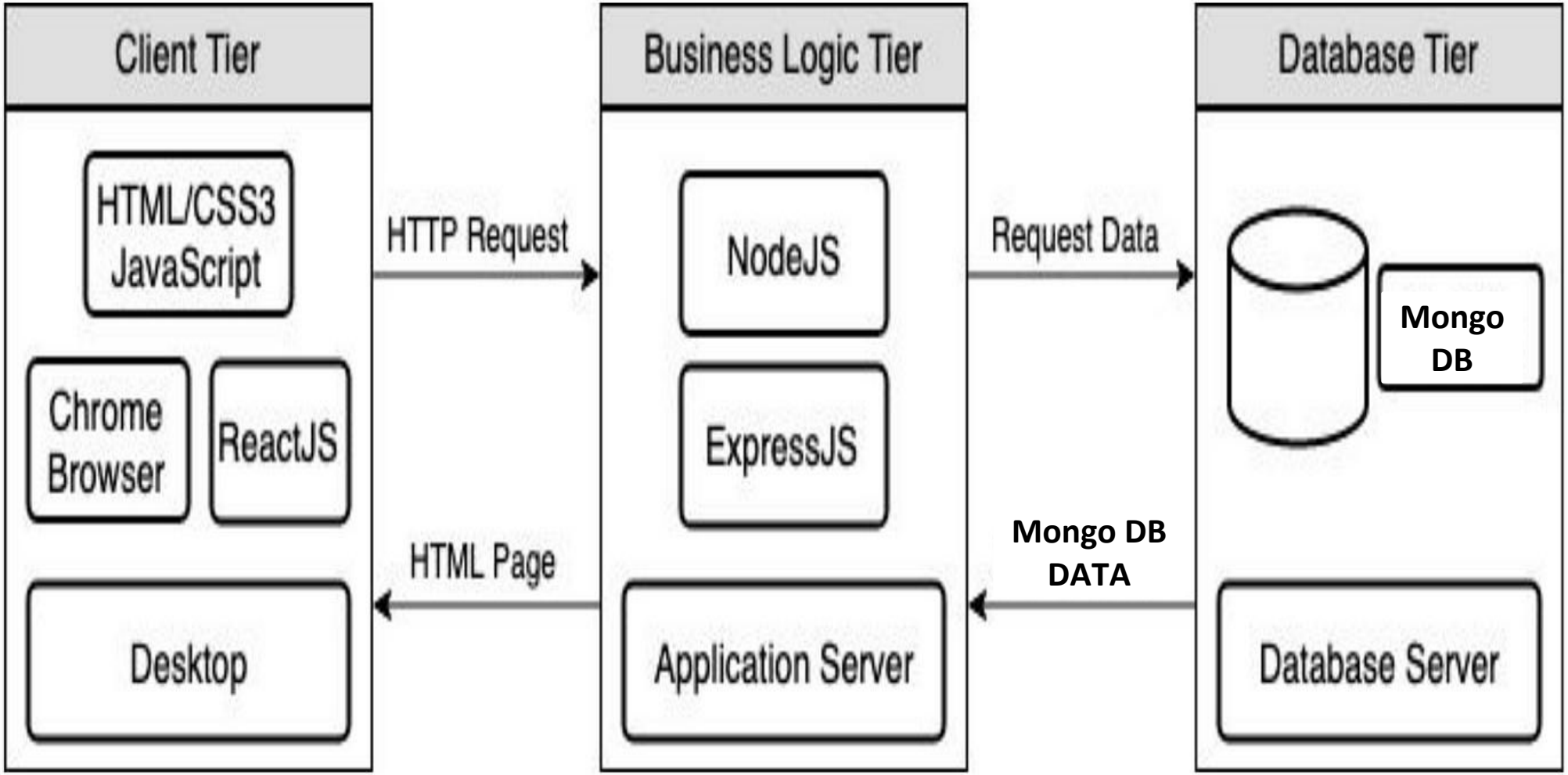
A JavaScript library for building user interfaces, allowing efficient and interactive web experiences.

- **Node.js**

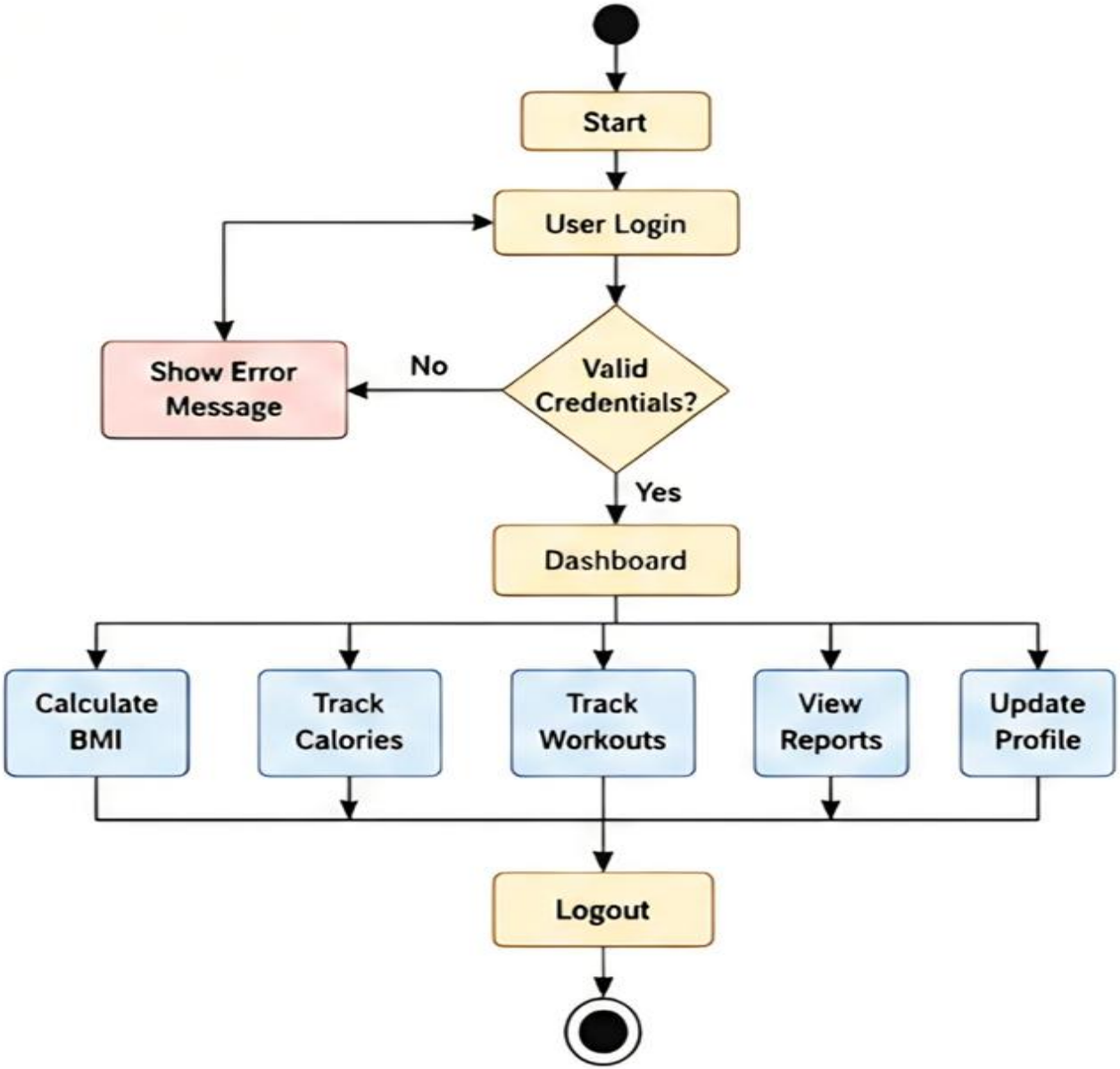
A JavaScript runtime environment, providing a server-side platform for executing JavaScript code.



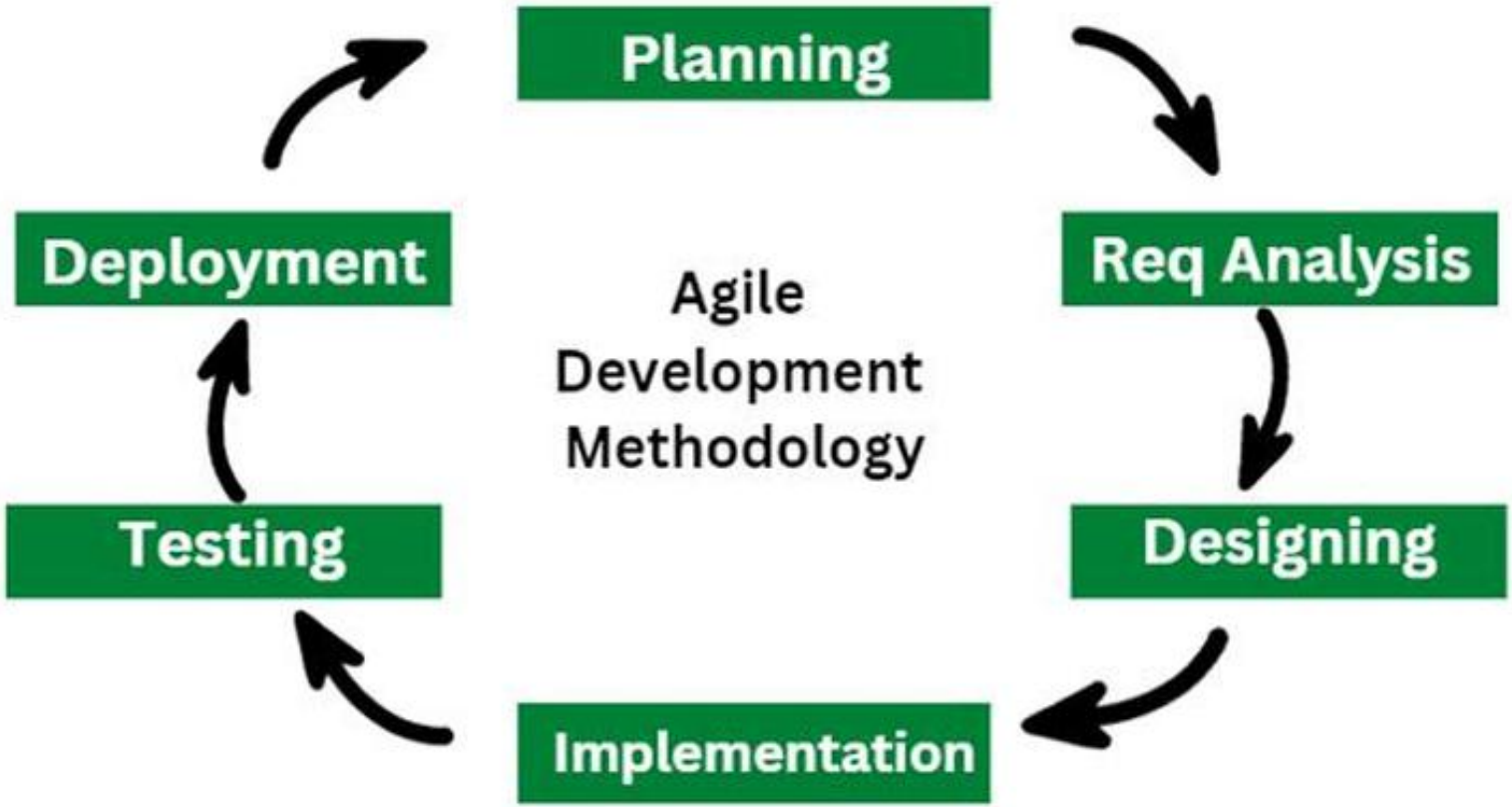
MERN Architecture



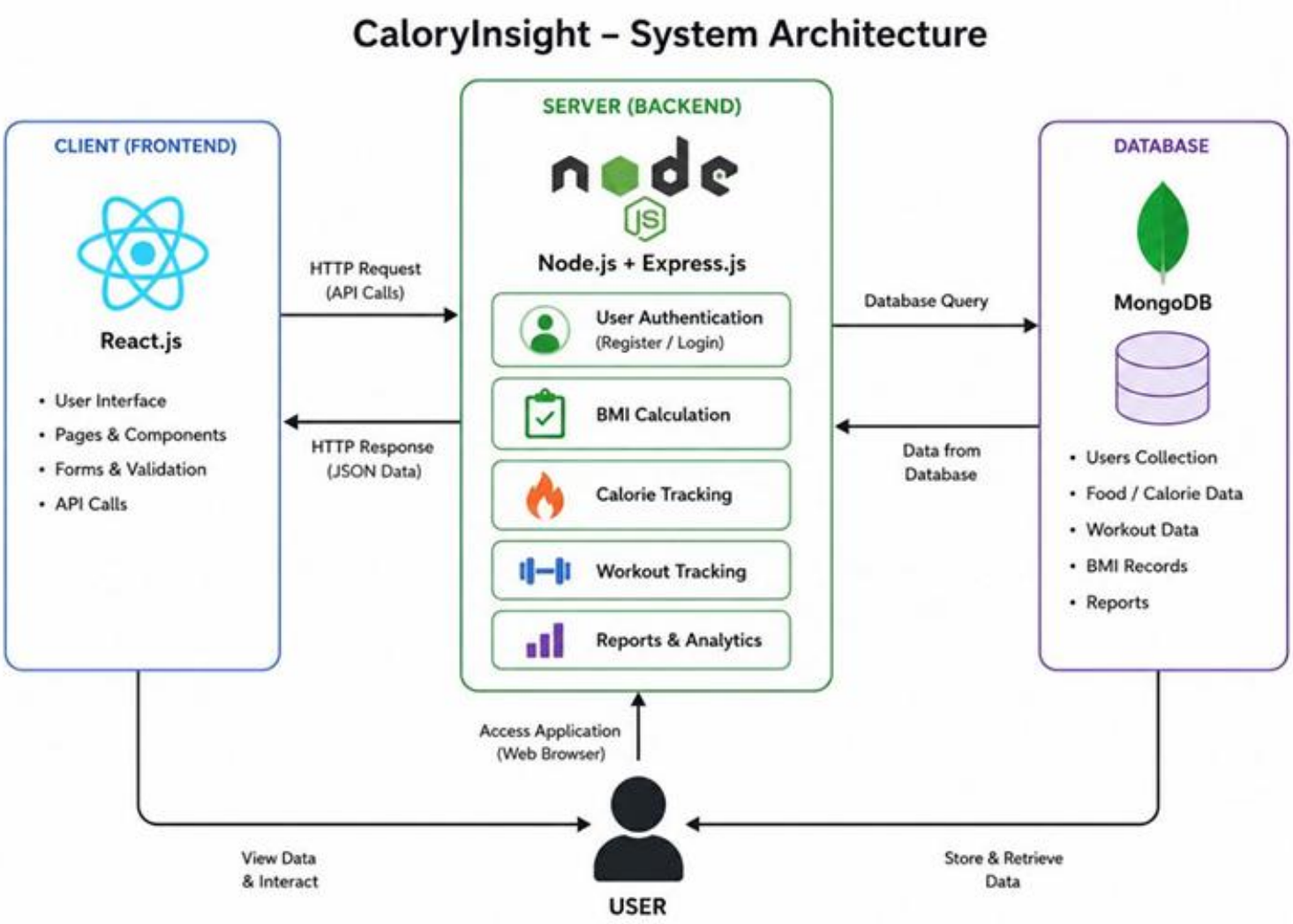
Data Flow Diagram



Agile Methodology



System Architecture



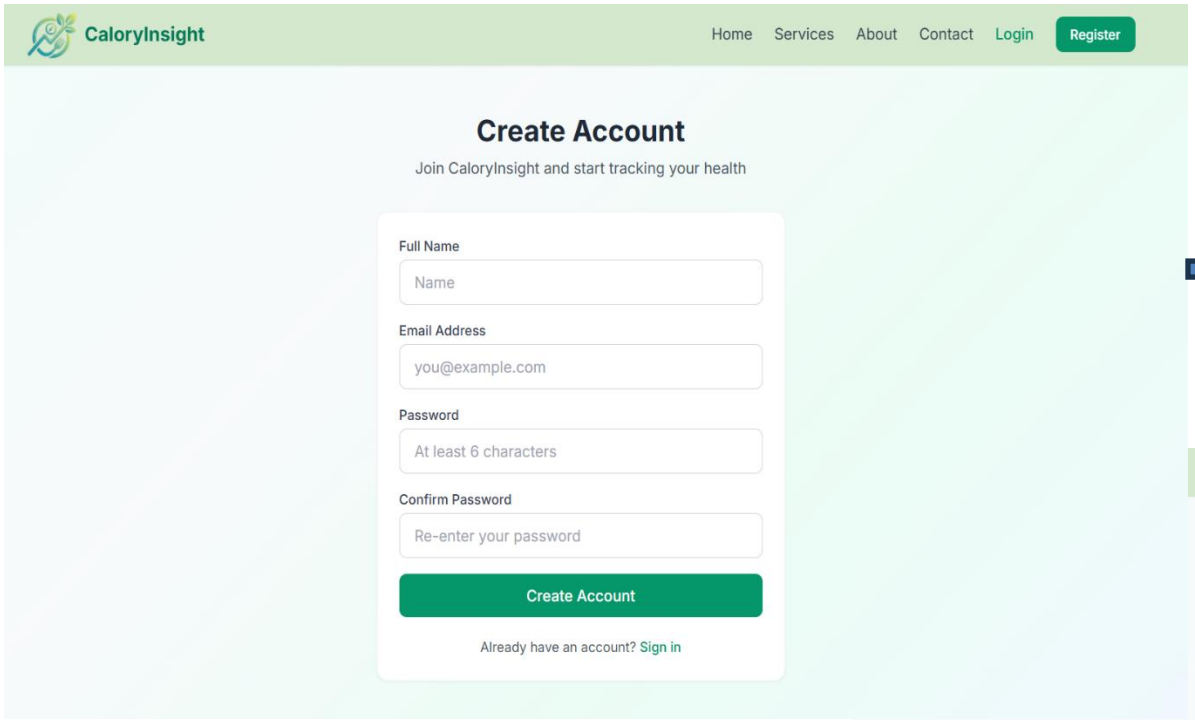
Outcomes

Sr.no	Outcomes	Status
1.	To Develop a user-friendly and secure LMS platform using MERN stack	In Progress
2.	To design a user-friendly interface	Completed
3.	To enable various Calorie-related services.	Completed
4.	To implement a contact form for users.	Completed
5.	To implement AI Tech Interface in platform.	In Progress
6.	To provide admin control for course and user management	Ongoing Testing

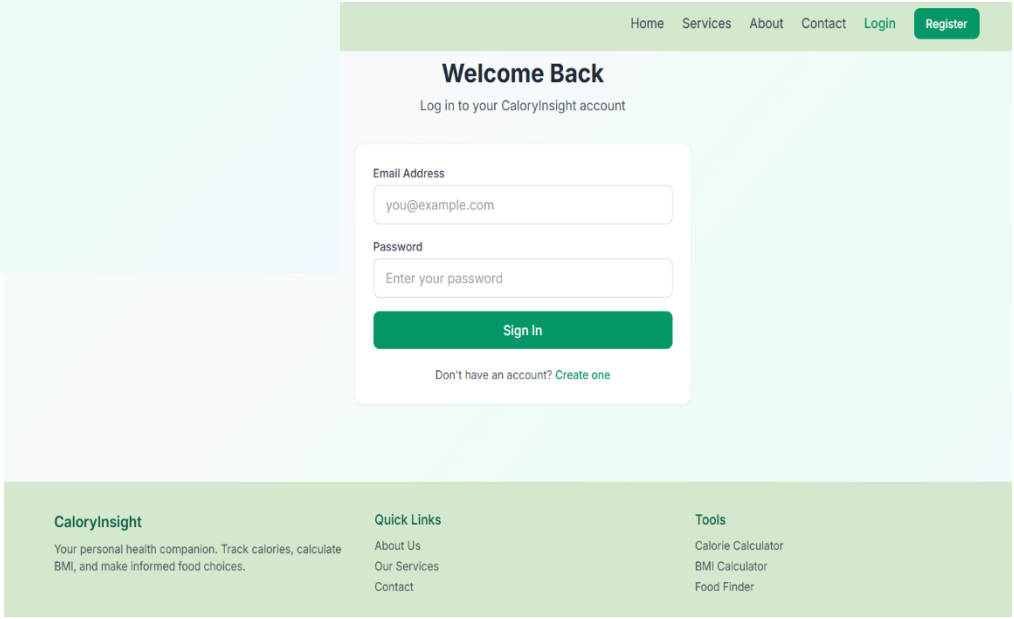
Outcome 1

Outcome1: To build a secure website

Description: The aim was to ensure the platform is safe for both admins and users by protecting user data and preventing unauthorized access.



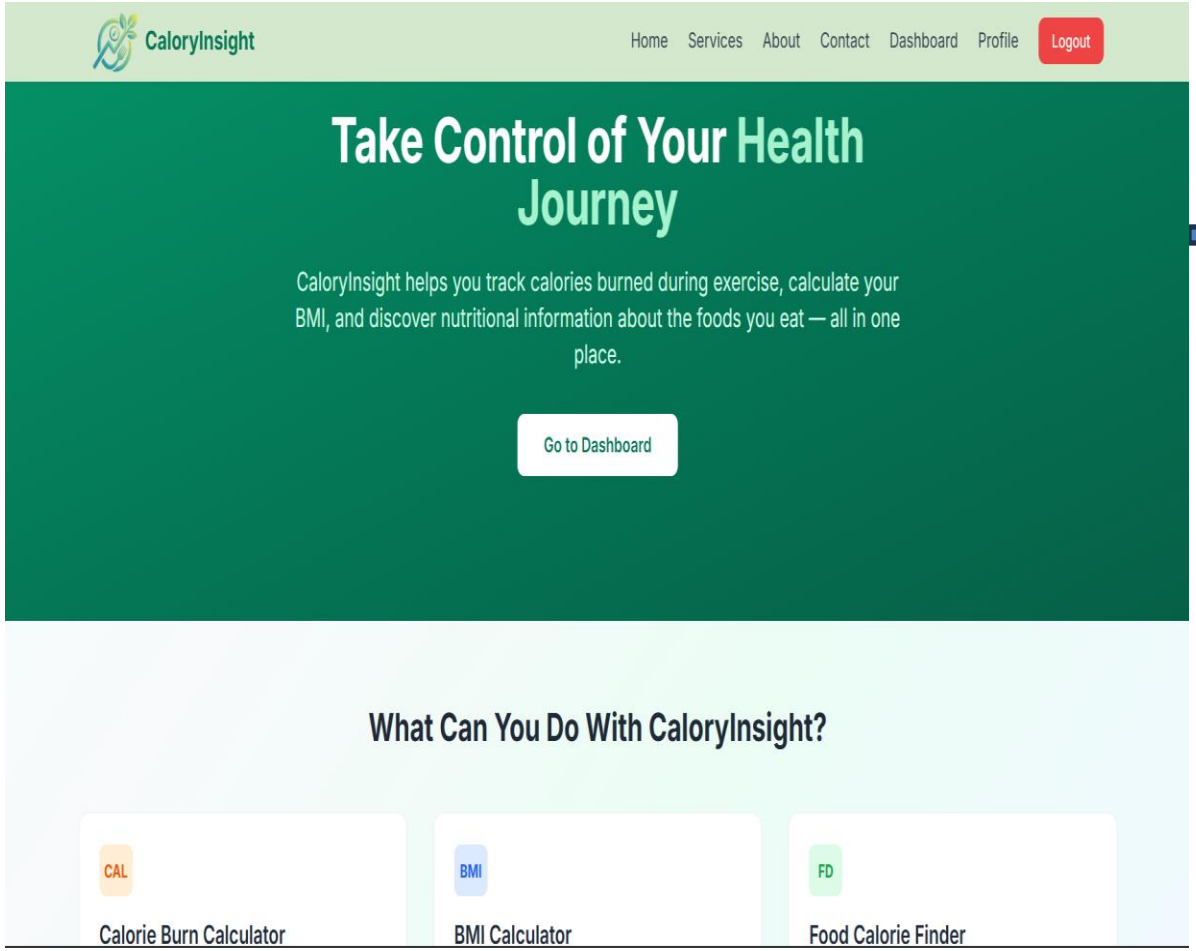
✨ **Feature Highlighted: Authentication and Authorization** — implemented using **JWT (JSON Web Tokens)** and **bcrypt.js** for password encryption.



Outcome 2

Outcome2: To design a user-friendly interface

Description: The goal was to create a smooth and intuitive user experience so that users and admins can easily navigate the platform.

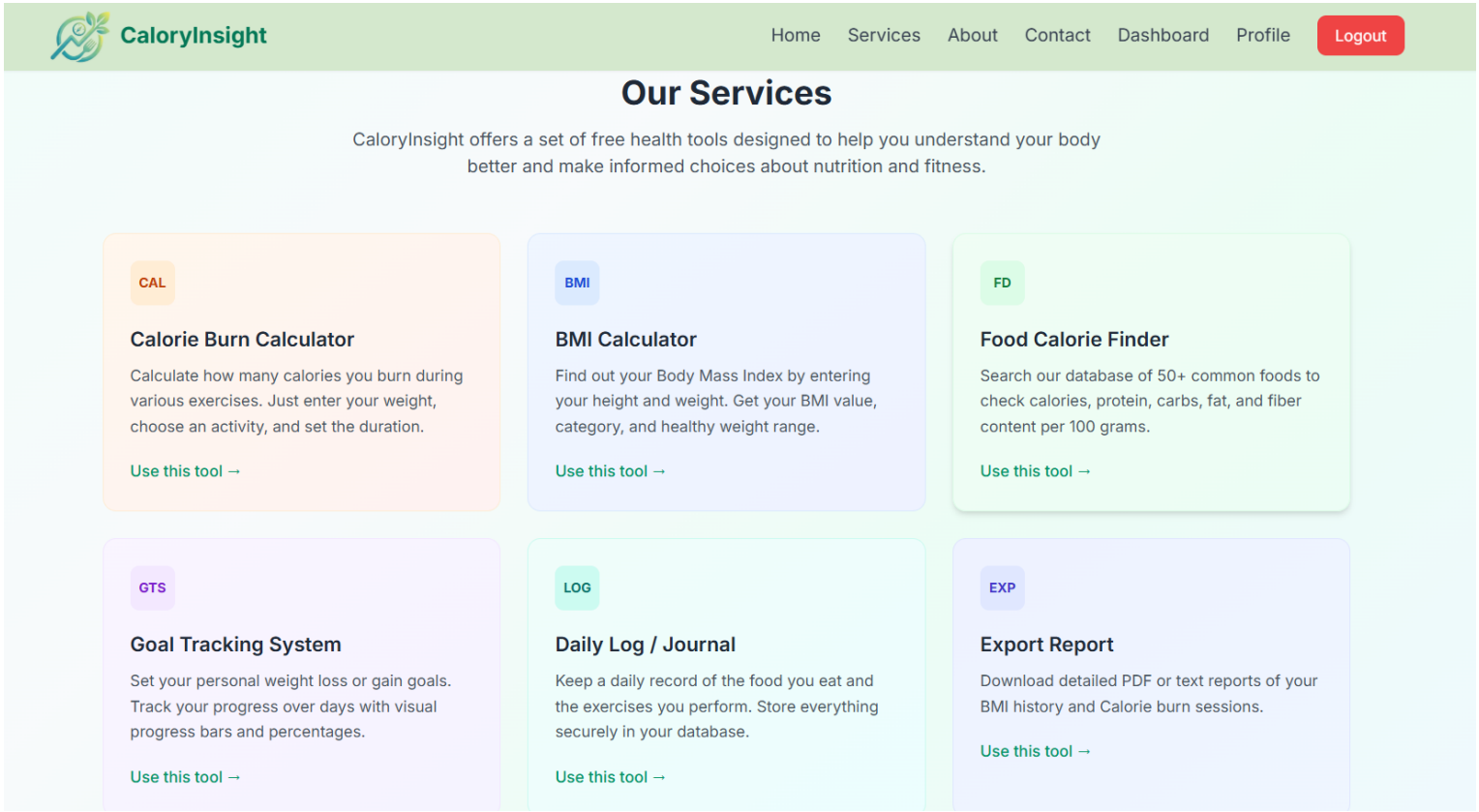


✨ **Feature Highlighted: Responsive UI Design** — developed using **React.js** and **Ant Design (antd)** for consistent and aesthetic layout across all devices.

Outcome 3

Outcome3: To enable various Calorie-related services.

Description: This objective focused on allowing users to empower our various calorie-related services with cutting-edge technologies used and personalized solutions.

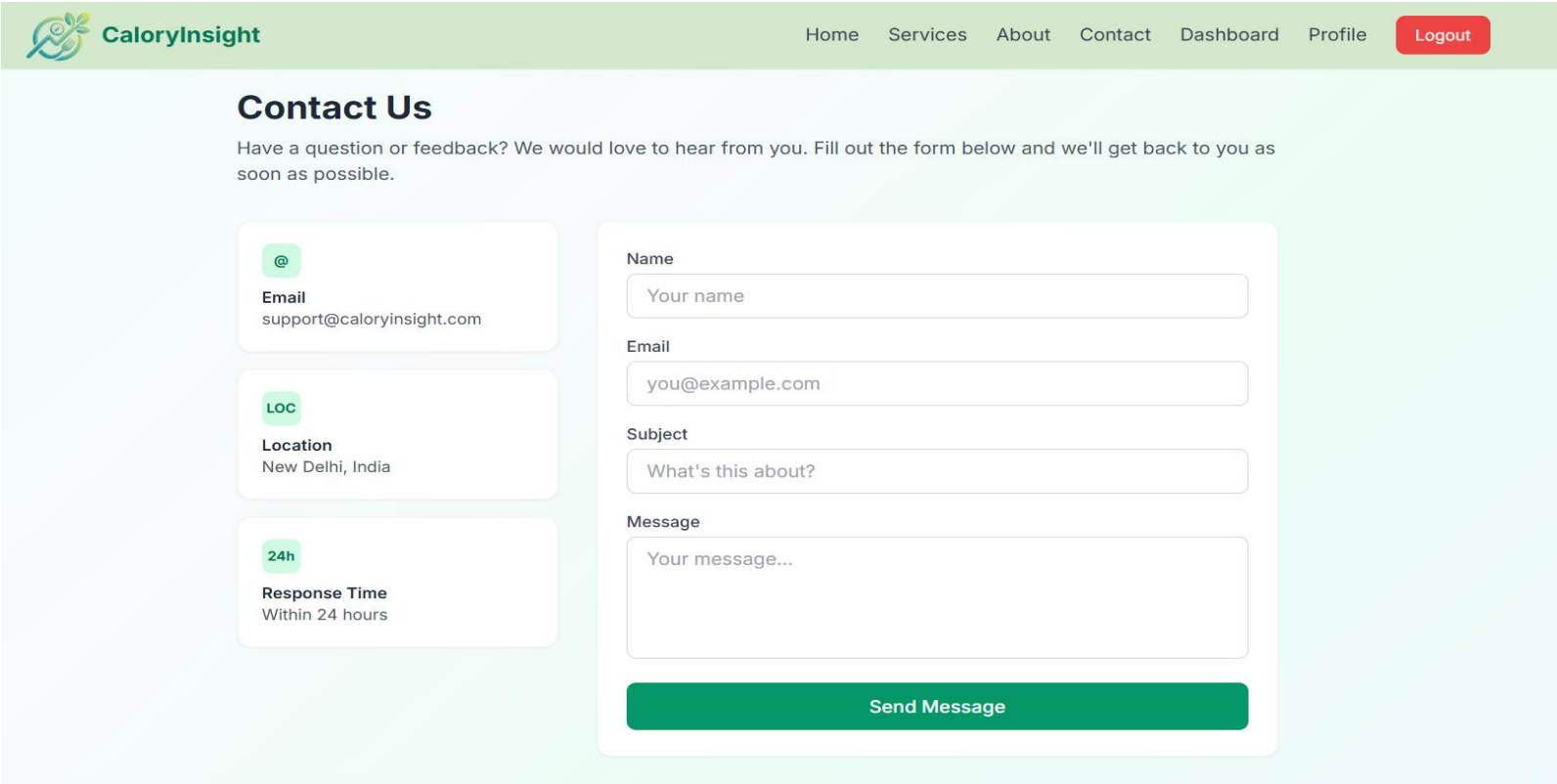


✨ **Feature Highlighted: Financial Services Integration (Test Mode)** — ensures safe and reliable use of various services.

Outcome 4

Outcome4: To implement a contact form for users.

Description: The forum allows users to post questions, share knowledge, and discuss topics, promoting community learning.



✨ **Feature Highlighted: Contact Us Feature** — built using **React.js** and **MongoDB** to store and display queries in real-time.

Conclusion &Future Enhancements

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•**CaloryInsight** successfully addresses the need for a simple and integrated health management system by combining multiple fitness tools into one platform.

•It enables users to monitor their calorie intake, calculate BMI, and track daily activities efficiently, helping them stay aware of their health status.

•The application's clean and user-friendly interface ensures that users can easily navigate and utilize its features without technical difficulty.

•By reducing dependency on multiple applications, it saves time and provides a centralized solution for fitness tracking.

•Overall, CaloryInsight acts as a reliable digital companion that motivates users to adopt healthier habits and maintain a balanced lifestyle.

Future Enhancements

- **Personalized Recommendations**

Provide diet plans and workout suggestions based on user data and goals.

- **AI-Based Insights**

Use machine learning to analyze user habits and give smarter health predictions.

- **Integration with Wearable Devices**

Connect with smartwatches or fitness bands for real-time activity tracking.

- **Advanced Analytics Dashboard**

Add graphs and reports to visualize progress (weekly/monthly trends).

- **Mobile Application Development**

Expand the platform into Android/iOS apps for better accessibility.

- **Social & Community Features**

Allow users to share progress, compete, and stay motivated with others.

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Thank You